TAGORE ENGINEERING COLLEGE

Rathinamangalam, Chennai-127.



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CS25C09 – JAVA PROGRAMMING LABORATORY

Name : _____

Reg.No : _____

Branch : _____

Year : _____

Semester : _____

Anna University: Chennai

TAGORE ENGINEERING COLLEGE

RATHINAMANGALAM, CHENNAI-600127

LABORATORY RECORD

UNIVERSITY REGISTER NO. | |

Certified that this is the bonafide record of work done by

Mr. /Ms._____of _____

Department in the_____Laboratory

and submitted for University Practical Examination conducted on _____

at TAGORE ENGINEERING COLLEGE, CHENNAI-127.

Record Marks _____.

Lab In-charge Principal Head of the Department

External Examiner Internal Examiner

VISION

To create an environment which is conducive to produce competent Artificial Intelligence and Data Science Engineers through quality education and research oriented education and equip them for the needs of the industry and society.

MISSION

The Department strives to contribute to the expansion of knowledge in the discipline of Artificial Intelligence and Data Science by

- Adopting an efficient teaching learning process in concurrence with increasing industrial demands.
- Ensuring technical proficiency, facilitating to pursue higher studies and carry out Research & Development activities.
- Developing problem solving and analytical skills with deep knowledge in thorough understanding of basic sciences and Artificial Intelligence and Data Science
- Infusing managerial and entrepreneurship skills to become ethical, socially responsible and competitive professionals.

PROGRAM SPECIFIC OBJECTIVES (PSOs)

1. Exhibit design and programming skills to build and automate business solutions using cutting edge technologies
2. Strong theoretical foundation leading to excellence and excitement towards research, to provide elegant solutions to complex problems.
3. Ability to work effectively with various engineering fields as a team to design, build and develop system applications.

PROGRAM OUTCOMES POs

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.

2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11. **Project management and finance**: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one‘s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological

COURSE OUTCOMES

CO	Description of CO	PO	PSO
CO1	Describe the object-oriented features of Java programming for developing robust applications.	--	--
CO2	Analyze object-oriented constructs to understand program flow, data handling, and exception management.	PO1 (3)	PSO1 (3)
CO3	Evaluate appropriate Java programming constructs in different scenarios.	PO3 (2)	PSO2 (2)
CO4	Design Java applications to solve real-world problems.	PO2 (1)	PSO1 (3)
CO5	Develop the ability to adapt to emerging Java frameworks and tools.	PO11 (2)	---

LIST OF EXPERIMENTS

Exp. No.	Date	Name of The Experiments	Page No.	Marks	Sign.

EX.NO.:1	Implementation of Constructors and Static Members
DATE:	

AIM:

To write a Java program that demonstrates the use of constructors and static members in a class.

ALGORITHM:

Step1: Create a class named Student.

Step2: Declare instance variables name and rollNo.

Step3: Declare a static variable college.

Step4: Create a constructor to initialize the instance variables.

Step5: Create a method to display student details.

Step6: In the main method, create objects using the constructor.

Step7: Call the display method for each object.

Step8: Display the student details along with the static member.

Step9: End the program.

PROGRAM:

```
class Student {
// Instance variables
String name;
int rollNo;

// Static variable (shared by all objects)
static String college = "ABC College";
```

```
// Constructor
Student(String name, int rollNo) {
this.name = name;
this.rollNo = rollNo;
}

// Static method
static void displayCollege() {
System.out.println("College Name: " + college);
}

// Instance method
void displayStudent() {
System.out.println("Name: " + name);
System.out.println("Roll No: " + rollNo);
System.out.println("College: " + college);
System.out.println();
}
}

public class ConstructorsAndStaticDemo {
public static void main(String[] args) {

// Calling static method
Student.displayCollege();

// Creating objects using constructor
```

```
Student s1 = new Student("Arun", 101);

Student s2 = new Student("Priya", 102);


// Display student details

s1.displayStudent();

s2.displayStudent();


// Changing static variable

Student.college = "XYZ University";


System.out.println("After changing static member:\n");


s1.displayStudent();

s2.displayStudent();

}

}
```

OUTPUT:

```
D:\Java\CS25C09>javacConstructor
AndStaticDemo.java
D:\Java\CS25C09>java
ConstructorsAndStaticDemo
```

College Name: ABC College

Name: Arun

Roll No: 101

College: ABC College

Name: Priya

Roll No: 102

College: ABC College

After changing static member:

Name: Arun

Roll No: 101

College: XYZ University

Name: Priya

Roll No: 102

College: XYZ Universit

RESULT:

Thus, the Java program demonstrating the use of constructors and static members was executed successfully. The constructor initialized the object data, and the static member was shared among all objects of the class.

EX.NO.: 2	**Develop a java application with an Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of BP as PF, 0.1% of BP for staff club funds. Generate pay slips for the employees with their gross and net salary**
DATE:	

AIM:

To develop a java application to generate pay slip for different category of employees using the concept of inheritance.

ALGORITHM:

Step 1: Start the program.

Step 2: Create the class Employee with name, Empid, address, mailid, mobileno as fields.

Step 3: Inherit the classes Programmer, AssistantProfessor, AssociateProfessor and Professor from employeeclass.

Step 4: Add Basic Pay (BP) as the member of all the inherited classes.

Step 5: Calculate DA as 97% of BP, HRA as 10% of BP, PF as 12% of BP, Staff club fund as 0.1%

Step 6: Calculate gross salary and net salary.

Grosssal=BP+HRA+DA

NetSal=GrossSal-(PF+Staff Club Fund)

Step 7:Create a test class PaySlip. Reademployee details, choice and basic pay from user. Step 8:Based on user choice and input create the object and invoke the necessary methods to display the Payslip.

Step 9: Stop the program.

PROGRAM:

PaySlip.java

```
import java.util.Scanner;
class Employee
{
   String Emp_name,Mail_id,Address,Emp_id, Mobile_no;
   double BP,GP,NP,DA,HRA,PF,CF;
   Scanner get = new Scanner(System.in);
Employee()
```

```
        System.out.println("Enter Name of the Employee:");
        Emp_name = get.nextLine();
        System.out.println("Enter Address of the Employee:");
        Address = get.nextLine();
        System.out.println("Enter ID of the Employee:");
        Emp_id = get.nextLine();
        System.out.println("Enter Mobile Number:");
        Mobile_no = get.nextLine();
        System.out.println("Enter E-Mail ID of the Employee :");
        Mail_id = get.nextLine();
    }
    void display()
    {
        System.out.println("Employee Name: "+Emp_name);
        System.out.println("Employee Address: "+Address);
        System.out.println("Employee ID: "+Emp_id);
        System.out.println("Employee Mobile Number: "+Mobile_no);
        System.out.println("Employee E-Mail ID"+Mail_id);
        DA=BP*0.97;
         HRA=BP*0.10;
         PF=BP*0.12;
         CF=BP*0.01;
         GP=BP+DA+HRA;
         NP=GP-PF-CF;
        System.out.println("Basic Pay :"+BP);
        System.out.println("Dearness Allowance : "+DA);
        System.out.println("House Rent Allowance :"+HRA);
        System.out.println("Provident Fund :"+PF);
        System.out.println("Club Fund :"+CF);
        System.out.println("Gross Pay :"+GP);
        System.out.println("Net Pay :"+NP);
    }
}
class Programmer extends Employee
{
        Programmer()
        {
                System.out.println("Enter Basic pay of the Programmer:");
                BP = get.nextFloat();
        }
        void display()
        {
        System.out.println("==============================="+"\n"+"Programmar
        Pay Slip"+"\n"+"==============================="+"\n");
        super.display();
        }
}
class AssistantProfessor extends Employee
{
        AssistantProfessor()
```

```
        {
                System.out.println("Enter Basic pay of the Assistant Professor:");
                BP = get.nextFloat();
         }
        void display()
        {
      System.out.println("================================="+"\n"+"Assistant ProfessorPay
Slip"+"\n"+"================================="+"\n");
super.display();
   }
}
class AssociateProfessor extends Employee
{
        AssociateProfessor()
        {
                System.out.println("Enter Basic pay of the Professor:");
                BP = get.nextFloat();
        }
        void display()
        {
         System.out.println("================================="+"\n"+"Associate
        Professor Pay Slip"+"\n"+"================================="+"\n");
        super.display();
        }
}
class Professor extends Employee
{
        Professor()
        {
                System.out.println("Enter Basic pay of the Professor:");
                BP = get.nextFloat();
         }
        void display()
        {
        System.out.println("================================="+"\n"+"Professor Pay
        Slip"+"\n"+"================================="+"\n");
        super.display();
        }
}
class Payslip
{
   public static void main(String[] args)
   {
        char ans;
        Scanner sc = new Scanner(System.in);
        do
        {
                System.out.println("Main Menu");
                System.out.println("1. Programmer \n2. Assistant Professor \n3. Associate
                Professor \n4. Professor");
```

```
                System.out.println("Enter your choice: ");
                int choice=sc.nextInt();
                switch(choice)
                {
                                case 1:
                                        Programmer p=new Programmer();
                                        p.display();
                                        break;
                                case 2:
                                        AssistantProfessorap=new AssistantProfessor();
                                        ap.display();
                                        break;
                                case 3:
                                        AssociateProfessor asp=new AssociateProfessor();
                                        asp.display();
                                        break;
                                case 4:
                                        Professor PR=new Professor();
                                        PR.display();
                                        break;
                }


                System.out.println("Do you want to go to Main Menu?(y/n): ");
                ans=sc.next().charAt(0);
                }while(ans=='y'||ans=='Y');
        sc.close();
    }
}
```

OUTPUT:

```
Main Menu
1. Programmer
2. Assistant Professor
3. Associate Professor
4. Professor
Enter your choice: 1
Enter Name of the Employee: Jeremiah
Enter Address of the Employee: Chennai
Enter ID of the Employee: 12345
Enter Mobile Number: 9360362173
Enter E-Mail ID of the Employee : jeremiahe@gmail.com
Enter Basic pay of the Programmer: 56000
==============================
Programmar Pay Slip
==============================
Employee Name: Jeremiah
Employee Address: Chennai
mployee ID: 12345
Employee Mobile Number: 9360362173
Employee E-Mail ID : jeremiahe@gmail.com
Basic Pay :56000.0
Dearness Allowance : 54320.0
House Rent Allowance :5600.0
Provident Fund :6720.0
Club Fund :560.0
Gross Pay :115920.0
Net Pay :108640.0
Do you want to go to Main Menu?(y/n): n
```

RESULT:

Thus, the Java application to generate pay slip for different category of employees was implemented using inheritance and the program was executed successful

EX.NO.: 3(a)	**Write a java program to create an abstract class named shape That contains two integers and an empty method named Printarea(). Provide three classes named rectangle, triangle and Circle such that each one of the classes extends the class shape. Each one of the classes contains only the method printarea( ) that Prints the area of the given shape. Solve the above problem using An interface**
DATE:	

AIM:

To write a Java program to calculate the area of rectangle, circle and triangle by implementing the interface shape.

ALGORITHM:

Step 1: Start the program.

Step 2: Create an interface named shape that contains an empty methodnamed printArea().

Step 3: Create three classes named rectangle, triangle and circle such that each one of the classes implements the class Shape.

Step 4: Each of the class should provide the implementation for the method printArea().

Step 5: In printAree() method get the input from user and calculate the area of rectangle, circle and triangle.

Step 6: In the TestAreaclass, create the objects for the three classes and invoke the MethodprintArea() and display the area values of the different shapes.

Step 7: Stop the program.

PROGRAM:

TestArea.java

```
import java.util.Scanner;
interface Shape
{
	public void printArea();
}
class Rectangle implements Shape
{
	public void printArea()
	{
		System.out.println("\t\tCalculating Area of Rectangle");
		Scanner input=new Scanner(System.in);
		System.out.print("Enter length: ");
		int a=input.nextInt();
		System.out.print("\nEnter breadth: ");
		int b=input.nextInt();
		int area=a*b;
		System.out.println("Area of Rectangle: "+area);
```

```
}
class Triangle implements Shape
{
	public void printArea()
	{
		System.out.println("\t\tCalculating Area of Triangle");
		Scanner input=new Scanner(System.in);
		System.out.print("Enter height: ");
		int a=input.nextInt();
		System.out.print("\nEnter breadth: ");
		int b=input.nextInt();
		double area=0.5*a*b;
		System.out.println("Area of Triangle: "+area);
	}
}
class Circle implements Shape
{
	public void printArea()
	{
		System.out.println("\t\tCalculating Area of Circle");
		Scanner input=new Scanner(System.in);
		System.out.print("Enter radius: ");
		int a=input.nextInt();
		double area=3.14*a*a;
		System.out.println("Area of Circle: "+area);
	}
}
public class TestArea
{
	public static void main(String[] args)
	{
		Shape obj;
		obj=new Rectangle();
		obj.printArea();
		obj=new Triangle();
		obj.printArea();
		obj=new Circle();
		obj.printArea();
	}
}
```

OUTPUT:

```
D:\Java\CS25C09>javacTestArea.jav

aD:\Java\CS25C09>java  TestArea

        Calculating Area of Rectangle
Enter length: 14

Enter breadth: 24
Area of Rectangle: 336
        Calculating Area of Triangle
Enter height: 45

Enter breadth: 32
Area of Triangle: 720.0
        Calculating Area of Circle
Enter radius: 5
Area of Circle: 78.5

D:\Java\CS25C09>
```

RESULT:

Thus, the Java program to calculate the area of rectangle, circle and triangle using the concept of interface was developed and executed successfully.

EX.NO.:3(b)	A PACKAGES WITH ACCESS SPECIFIERS
DATE:	

AIM:

To write a Java program that demonstrates the use of Packages**,** Interfaces**,** and Access Specifiers (public) in Java.

ALGORITHM:

Step1: Create a package named mypackage.

Step2: Inside the package, create a public interface named Shape.

Step3: Declare a method area() in the interface.

Step4: Create a public class Circle that implements the Shape interface.

Step5: Define the area() method in the Circle class.

Step6: Create another class PackageDemo outside the package.

Step7: Import the package.

Step8: Create an object of the Circle class.

Step9: Call the area() method using the object.

Step10: Display the result and terminate the program.

PROGRAM

File 1: Shape.java

```
package mypackage;

public interface Shape {
   void area();
}
```

File 2: Circle.java

```
package mypackage;

public class Circle implements Shape {
   double radius = 5;

   public void area() {
      double a = 3.14 * radius * radius;
      System.out.println("Area of Circle = " + a);
   }
}
```

File 3: PackageDemo.java

```
import mypackage.Circle;

public class PackageDemo {
   public static void main(String[] args) {

      Circle c = new Circle();
      c.area();
   }
}
```

OUTPUT:

D:\Java\CS25C09>javac PackageDemo.java

D:\Java\CS25C09>javaPackageDemo

Area of Circle = 78.5

RESULT:

Thus, the Java program demonstrating Packages**,** Interfaces**,** and Access Specifiers was executed successfully. The package mypackage was created, the interface was implemented and the public access specifier allowed the class and method to be accessed from another package

EX.NO.:4(a)	IMPLEMENT EXCEPTION HANDLING AND CREATION OF USER DEFINED EXCEPTIONS
DATE:	

AIM:

To write a Java program to implement user defined exception handling.

ALGORITHM:

Step 1: Start the program.

Step 2: Create a class NegativeAmtException which extends Exception class.

Step 3: Create a constructor which receives the string as argument.

Step 4: Create a class named BankAccount with a constructor and methods such as deposit(), withdraw(), getBalance() and toString().

Step 5: Inside the constructor, deposit() and withdraw() methods check for negative amount. If the amount is negative, then exception will be generated and thrown. Make these methods to specify that NegativeAmtException will be thrown.

Step 6: Create a test class UserDefinedException . Read the account number and initial balance from user and create object of BankAccount class.

Step 7: Display the menu and get the user choice and execute the required operation. Write the code that generates exception, in try block.

Step 8: Use catch block to catch and handle NegativeAmtException. Display the caught exception.

Step 9: Stop the program.

PROGRAM:

UserDefinedException.java

```
import java.util.*;
class NegativeAmtException extends Exception
{
        String msg;
        NegativeAmtException(String msg)
        {
                this.msg=msg;
        }
        public String toString()
        {
                return msg;
        }
}
```

```
class BankAccount
{
	private double balance;
	private intaccountNumber;
	public BankAccount(intaccountNumber,doubleinitialBalance)throws
	NegativeAmtException
	{
		if(initialBalance<0)
			throw new NegativeAmtException("Initial amount should not be
			negative!");
		balance = initialBalance;
		this.accountNumber = accountNumber;
	}
	public void deposit(double amount)throws NegativeAmtException
	{
		if (amount < 0)
		{
			throw new NegativeAmtException("Don't deposit negative amount!");
		}
		balance = balance + amount;
		System.out.println("Amount deposited");
		System.out.println("Balance amount : "+getBalance());
	}
	public void withdraw(double amount)throws NegativeAmtException
	{
		if (amount < 0)
		{
		throw new NegativeAmtException("Don't withdraw a negative
			amount!");
		}
		else if(amount<=balance)
		{
			balance = balance – amount;
		}
		else
		{
			System.out.println("Insufficient  amount");
		}


		System.out.println("Balance amount : "+getBalance());
	}
	public double getBalance()
	{
		return balance;
	}
	public intgetAccountNumber()
	{
		return accountNumber;
	}
	public String toString ()
	{
```

```
                return "Account Number :" + accountNumber + " Balance :" + balance;
        }
}
public class UserDefinedException
{
        public static void main(String[] args)
        {
                intch,amt;
                Scanner sc=new Scanner(System.in);
                System.out.print("Enter Account Number:");
                int a=sc.nextInt();
                System.out.print("Enter the initial Amount:");
                int b=sc.nextInt();
                BankAccount ac;
                try
                {
                        ac=new BankAccount(a,b);
                        while(true)
                        {
                                System.out.println("Main Menu");
                                System.out.println("1.Deposit \n2.Withdraw \n3.Check Balance
                                \n4.Display \n5.Exit");
                                System.out.print("Enter your Choice: ");
                                ch=sc.nextInt();
                                switch(ch)
                                {
                                        case 1:
                                                System.out.print("Enter the amount to deposit:");
                                                amt=sc.nextInt();
                                                ac.deposit(amt);
                                                break;
                                        case 2:
                                                System.out.print("Enter the amount to
                                                Withdraw:");
                                                amt=sc.nextInt();
                                                ac.withdraw(amt);
                                                break;
                                        case 3:
                                                System.out.println("Balance amount :
                                                "+ac.getBalance());
                                                break;
                                        case 4:



                                                System.out.println("Your account
                                                details\n"+ac);
                                                break;
                                        case 5:
                                                sc.close();
                                                System.exit(0);
                                         efault:
```

```
                                        System.out.println("Invalid Choice");
                                }
                        }
                }
                catch(NegativeAmtException e)
                {
                        System.out.println("Exception Caught : "+e);
                }
        }
}
```

OUTPUT:

```
D:\Java\CS25C09>javac.java

D:\Java\CS25C09>java



Enter Account Number:1234
Enter the initial Amount:500
Main Menu
1. Deposit
2. Withdraw
3. Check Balance
4. Display
5. Exit
Enter your Choice: 1
Enter the amount to deposit:-456
Exception Caught : Don't deposit negative amount!
```

RESULT:

Thus the Java program to implement user defined exception handling was implementedand executed successfully.

EX.NO.: 4 (b)	WRITE A JAVA PROGRAM THAT IMPLEMENTS A MULTI-THREADED APPLICATION THAT HAS THREE THREADS. FIRST THREAD GENERATES A RANDOM INTEGER EVERY 1 SECOND AND IF THE VALUE IS EVEN, THE SECOND THREAD COMPUTES THE SQUARE OF THE NUMBER AND PRINTS. IF THE VALUE IS ODD, THE THIRD THREAD WILL PRINT THE VALUE OF THE CUBE OF THE NUMBER
DATE:	

AIM:

To write a java program to implement a multi-threaded application.

ALGORITHM:

Step 1: Start the program.
Step 2: Create a class even which implements first thread that computes the square of the number.
Step 3:Therun() method implements the code to be executed when thread gets executed.
Step 4:Create a class odd which implements second thread that computes the cube of the number.
Step 5:Create a third thread that generates random number. If the random number is even, it displaysthe square of the number. If the random number generated is odd, it displays the cube of thegiven number.
Step 6: The Multithreading is performed and the task switched between multiple threads.
Step 7: The sleep () method makes the thread to suspend for the specified time.
Step 8: Stop the program.

PROGRAM:

MultiThread.java

```
import java.util.*;
class even implements Runnable
{
        public int x;
        public even(int x)
        {
                this.x = x;
        }
        public void run()
        {
                System.out.println("New Thread "+ x +" is EVEN and Square of " + x + " is: "
                + x * x);
        }
}
```

```
class odd implements Runnable
{
	public int x;
	public odd(int x)
	{
		this.x = x;
	}
	public void run()
	{
		System.out.println("New Thread "+ x +" is ODD and Cube of " + x + " is: " +
		x * x * x);
	}
}
class A extends Thread
{
	public void run()
	{
		intnum = 0;
		Random r = new Random();
		try
		{
			for (inti = 0; i< 5; i++)
			{
				num = r.nextInt(100);
				System.out.println("Main Thread and Generated Number is " +
				num);
				if (num % 2 == 0)
				{
					Thread t1 = new Thread(new even(num));
					t1.start();
				}
				else
				{
					Thread t2 = new Thread(new odd(num));
					t2.start();
				}


				Thread.sleep(1000);
				System.out.println(" ----------------------------------------------");
			}
		}
		catch (Exception ex)
		{
			System.out.println(ex.getMessage());}
```

```
        }
}
public classMultiThread
{
        public static void main(String[] args)
        {
                A a = new A();
                a.start();
        }
}
```

OUTPUT:

```
Main Thread and Generated Number is 33
New Thread 33 is ODD and Cube of 33 is: 35937
---------------------------------------------------
Main Thread and Generated Number is 31
New Thread 31 is ODD and Cube of 31 is: 29791
---------------------------------------------------
Main Thread and Generated Number is 25
New Thread 25 is ODD and Cube of 25 is: 15625
---------------------------------------------------
Main Thread and Generated Number is 43
New Thread 43 is ODD and Cube of 43 is: 79507
---------------------------------------------------
Main Thread and Generated Number is 14
New Thread 14 is EVEN and Square of 14 is: 196
---------------------------------------------------
```

RESULT:

Thus, the java program to implement multithreaded application was executed successfully.

EX.NO.: 5	STRING HANDLING
DATE:	

AIM

To write a Java program to demonstrate string handling operations such as length, concatenation, character extraction, comparison, and case conversion

ALGORITHM

Step1: Create a Java class.

Step2: Declare and initialize string variables.

Step3: Perform string operations such as finding length of a string, Concatenation of strings, converting to uppercase and lowercase, extracting a character, Comparing two strings

Step4: Display the results of each operation.

Step5:End the program.

PROGRAM:

```
public class StringHandlingDemo {
   public static void main(String[] args) {

      // String initialization
      String str1 = "Java";
      String str2 = "Programming";

      // Length of string
      System.out.println("Length of str1: " + str1.length());

      // Concatenation
      System.out.println("Concatenation: " + str1.concat(str2));

      // Uppercase and Lowercase
      System.out.println("Uppercase: " + str1.toUpperCase());
      System.out.println("Lowercase: " + str2.toLowerCase());

      // Character extraction
      System.out.println("Character at index 2 of str1: " + str1.charAt(2));
```

```
        // String comparison
        System.out.println("Comparison: " + str1.equals(str2));

        // Substring
        System.out.println("Substring of str2 (0-4): " + str2.substring(0, 4));
    }
}
```

OUTPUT:

Length of str1: 4
Concatenation: JavaProgramming
Uppercase: JAVA
Lowercase: programming
Character at index 2 of str1: v
Comparison: false
Substring of str2 (0-4): Prog

RESULT:

Thus, the Java program demonstrating string handling operations was successfully executed. Various string methods such as length (), concat(), toUpperCase(), toLowerCase(), charAt(), equals(), and substring() were used to manipulate and analyze strings

EX.NO:6	Event Handling
DATE:	

AIM:

To study and implement event handling in Java using the ActionListener interface and handle a button click event.

ALGORITHM:

Step1: Import the required packages java.awt.* and java.awt.event.*.
Step2: Create a class that extends Frame and implements ActionListener.
Step3: Create a Label and a Button.
Step4: Add the button and label to the frame.
Step5: Register the button with the event listener using addActionListener().
Step6: Implement the actionPerformed() method.
Step7: When the button is clicked, change the label text.
Step8: Display the frame and execute the program.

PROGRAM

```
import java.awt.*;
import java.awt.event.*;

public class EventHandlingDemo extends Frame implements ActionListener {

    Label lbl;
    Button btn;

    EventHandlingDemo() {

        lbl = new Label("Click the Button");
        lbl.setBounds(50, 50, 150, 30);

        btn = new Button("Click Me");
        btn.setBounds(50, 100, 80, 30);

        btn.addActionListener(this);

        add(lbl);
        add(btn);

        setSize(300, 200);
        setLayout(null);
```

```
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        lbl.setText("Button Clicked!");
    }

    public static void main(String[] args) {
        new EventHandlingDemo();
    }
}
```

OUTPUT:

Before Clicking the Button

```
Click the Button

[ Click Me ]
```

After Clicking the Button

```
Button Clicked!

[ Click Me ]
```

RESULT:

Thus, the Java program for **event handling using the ActionListener interface** was implemented and executed successfully. The button click event was handled, and the label text changed when the button was clicked.

EX.NO.: 7	FILE OPERATIONS
DATE:	

AIM:

To write a java program to copy the contents of one file to another file using file operations.

ALGORITHM:

Step 1: Start the program.

Step 2: Create a class FileCopy. Get the source and destination file names from the user.

Step 3: Create object of FileInputStream by passing the source file name.

Step 4: Using read() method read the contents of the file till end of the file.

Step 5: Create object of FileOutputStream by passing the second file to the constructor.

Step 6: Use write() method to write the contents to the destination file.

Step 7: Close the file input and output stream using close() method.

Step 8: Display the contents of both files.

Step 9: Stop the program.

PROGRAM:

FileCopy.java

```
import java.io.*;
class CopyFile
{
        public static void main(String args[]) throws IOException
        {
                inti;
                FileInputStream fin = null;
                FileOutputStreamfout = null;
                if(args.length != 2)
                {
                        System.out.println("Usage: CopyFile from to");
                        return;
                }
                System.out.println("Displaying contents of "+ args[0]+"\n");
                try
                {
                        fin = new FileInputStream(args[0]);
```

```
        {
                i = fin.read();
                if(i != -1)
                        System.out.print((char) i);
        } while(i != -1);
}
catch(IOException e)
{
        System.out.println("Error Reading File");
}
finally
{
        try
        {
                fin.close();
        }
        catch(IOException e)
        {
                System.out.println("Error Closing File");
        }
}
System.out.println("\nCopying contents of "+ args[0] + "to " + args[1]+"\n");
try
{
        fin = new FileInputStream(args[0]);
        fout = new FileOutputStream(args[1]);
        do
        {
                i = fin.read();
                if(i != -1) fout.write(i);
        } while(i != -1);
}
catch(IOException e)
{
        System.out.println("I/O Error: " + e);
}
finally
{
        try
        {
                if(fin != null)
                        fin.close();
        }
        catch(IOException e2)
```

```
                {
                        System.out.println("Error Closing Input File");
                }
                try
                {
                        if(fout != null)
                                fout.close();
                }


                catch(IOException e2)
                {
                        System.out.println("Error Closing Output File");
                }
        }
        System.out.println("\nFile Copied\n");
        System.out.println("\nDisplaying contents of "+ args[1]+"\n");
        try
        {
                fin = new FileInputStream(args[1]);
                do
                {
                        i = fin.read();
                        if(i != -1)
                                System.out.print((char) i);
                } while(i != -1);
        }
        catch(IOException e)
        {
                System.out.println("Error Reading File");
        }
        finally
        {
                try
                {
                        fin.close();
                }
                catch(IOException e)
                {
                        System.out.println("Error Closing File");
                }
        }
    }
}
```

OUTPUT:

```
R: javac CopyFile.java
R: java  CopyFile FIRST.txt SECOND.txt

Displaying contents of FIRST.txt

To use this program, specify the name
of the source file and the destination file.
For example, to copy a file called
FIRST.TXT  to a file called SECOND.TXT,
use the following command line.
java CopyFile FIRST.TXT SECOND.TXT

Copying contents of FIRST.txt to

SECOND.txt File Copied

Displaying contents of SECOND.txt

To use this program, specify the name
of the source file and the destination file.
For example, to copy a file called
FIRST.TXT  to a file called SECOND.TXT,
use the following command line.
java CopyFile FIRST.TXT

SECOND.TXT R:\oop>
```

RESULT:

Thus, the java program to copy the contents of one file to another file using file was written, executed and verified.

EX.N0 :7 (a)	
DATE:	**FILE OPERATIONS** **FILE CONTENT DISPLAY**

AIM

To write a java program to read the content from the file.

ALGORITHM

Step 1: Start.
Step 2: Create a file(hello.txt) with the content for reading.
Step 3: Create file input stream object fis from read hello.txt
Step 4: Inside try block create a variable c to display the character.
Step 5: While((c=fis.read())!=-1) then display the content of the file.
Step 6: If(fis!=NULL),then close the file by fis.close().
Step 7: Stop.

PROGRAM

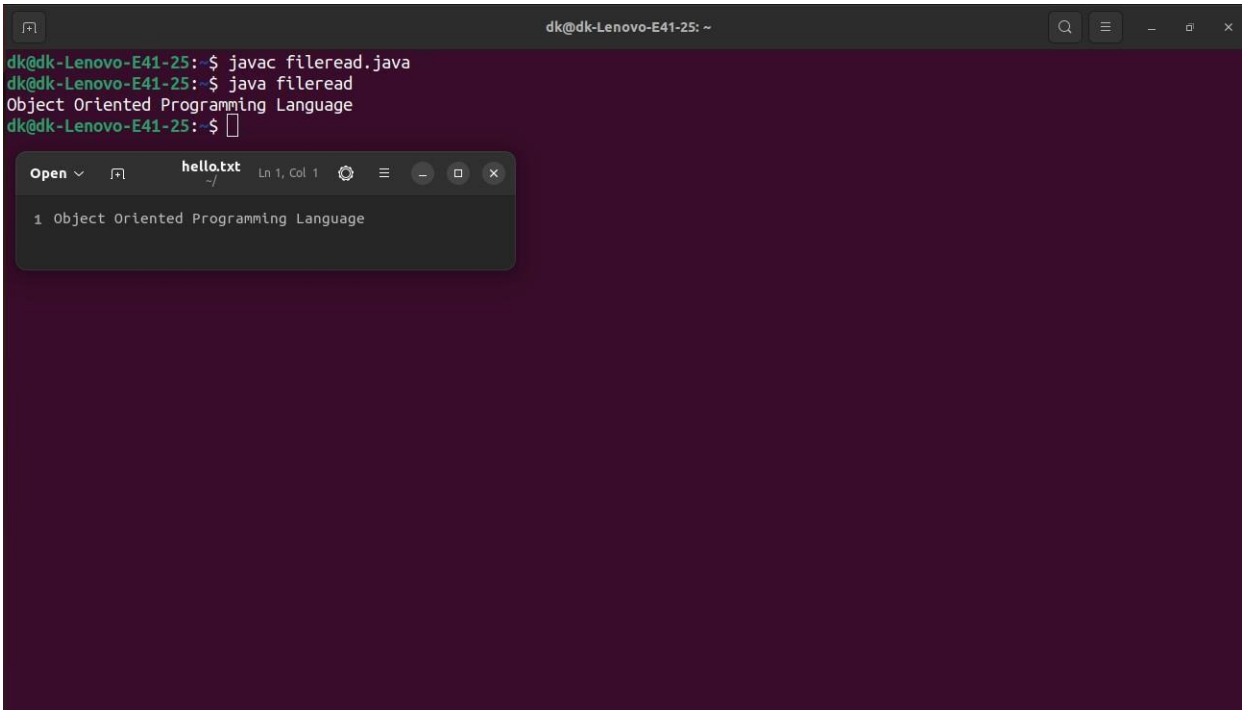
```
import java.io.*; class fileread
{
        public static void main(String args[]) throws IOException
        {
                FileInputStream fis=new FileInputStream("hello.txt");
                try
                {
                        int c;
                        while((c=fis.read())!=-1)
                        {
                                System.out.print((char)c);
                        }

                }

                finally
                {
                        if(fis!=null)
                                fis.close();
```

```
        }
    }
}
```

OUTPUT



RESULT

Thus, the java program for read content from file has been executed successful

EX.NO:7(b)	FILE COPY
DATE:	

AIM

Write a java program to copy the content of one file to another file.

ALGORITHM:

Step 1: Start.

Step 2: Create two files one file with content and the other file to copy the content.

Step 3: Create file input stream object fis from read hello.txt and Create file input stream object fos to write hello.txt ("output.txt")

Step 4: Inside try block create a variable c to display the character.

Step 5: WHILE((c=fis.read())!=-1) then copy the content to the another file using fos.write(c)

Step 6: IF(fis!=NULL),close the file by fis.close().

Step 7: IF(fos!=null),close the file by fos.close().

Step 8: Stop.

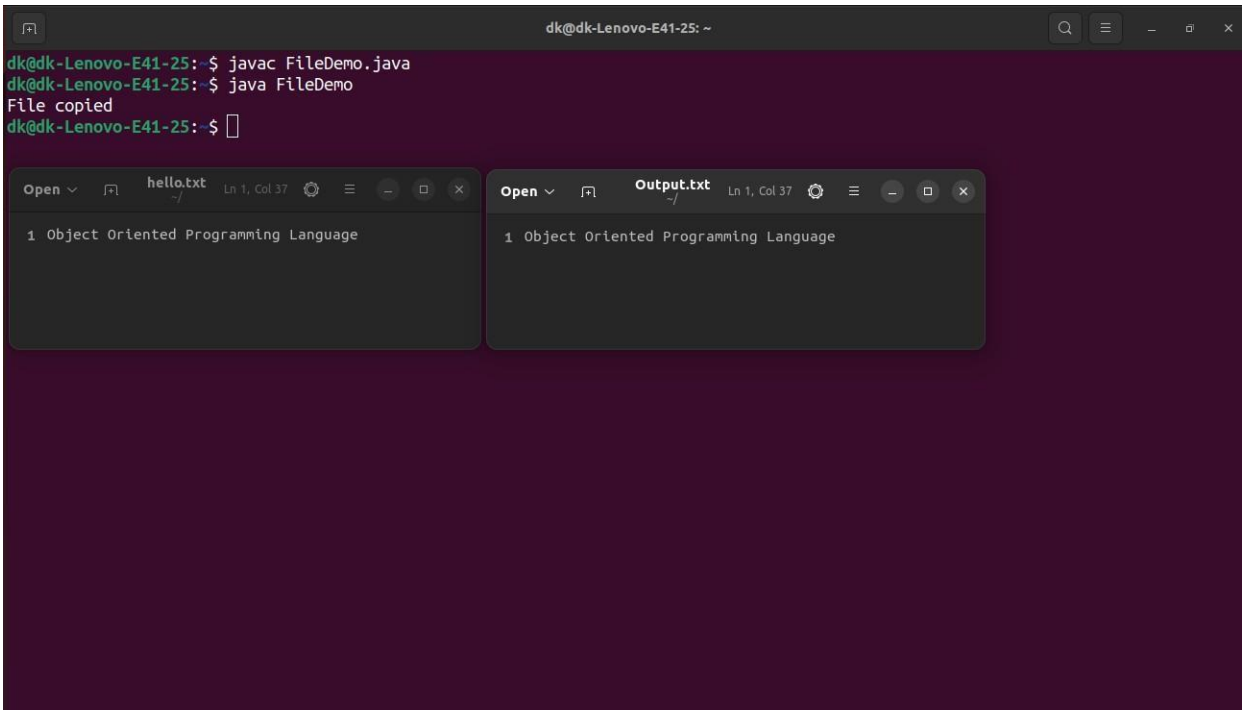
PROGRAM:

```
import java.io.*; class FileDemo
{
        public static void main(String args[]) throws IOException
        {
                FileInputStream fis=new FileInputStream("hello.txt");
                FileOutputStream fos =new FileOutputStream("Output.txt");
                try
                {

                        int c;
                        while((c=fis.read())!=-1)
                        {
                                fos.write(c);
                        }
```

```
                }
                finally
                {
                        if(fis!=null)
                                fis.close();
                        if(fos!=null)
                                fos.close();
                        System.out.println("File copied");
                }
        }
}
```

OUTPUT



RESULT

Thus, the java program for file copy has been executed successfully.

EX.NO:7 (C)	REVERSE FILE
DATE:	

AIM

To write a java program to reverse the text content of a file.

ALGORITHM

Step 1:Start.

Step 2: import required packages.

Step 3: Open a new or already existing file in read mode.

Step 4: Create an output file.

Step 5: Create an object fis and fos for input and output streams.

Step 6:Using the reverse(),reverse the content in the file by FOR loop.

Step 7: Close the files after the operations.

Step 8: Stop.

PROGRAM

```
import java.io.*;
import java.util.*;
class Main
{
	public static void main(String args[]) throws IOException
	{
		FileInputStream fis= new FileInputStream("DK.txt");
		FileOutputStream fos = new FileOutputStream("PK.txt");
		try
		{
			int ch;
			String str="",das="";
			while((ch=fis.read())!=-1)
			{
				str+=(char)ch;
			}
			StringBuilder i1 = new StringBuilder();
			i1.append(str);
```

```
System.out.println("File reversed successfully");



if(fis!=null)
        fis.close(); if(fos!=null)
         fos.close();
i1.reverse(); das=i1.toString();

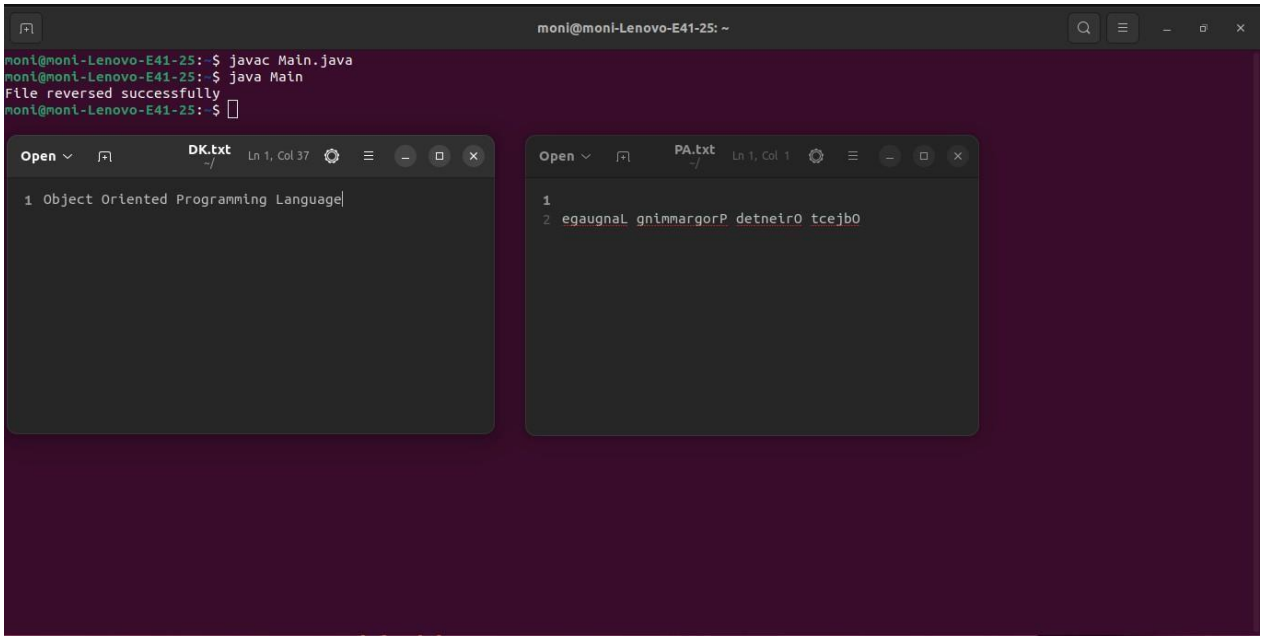
                        for(int i=0;i<das.length();i++)
                        {
                                fos.write(das.charAt(i));
                        }

                }
                finally
                {




                }
        }
}
```

OUTPUT



RESULT

Thus, the java program for file reverse has been executed successfully.

EX.NO:7(d)	FILE CONTENT CASE CONVERSION
DATE:	

AIM

To write a java program to convert the content of the file lowercase to uppercase.

ALGORITHM

Step 1: Start.

Step 2: Import required packages.

Step 3: Open a new or already existing file in read mode.

Step 4: Create an output file.

Step 5: Create an object fis and fos for input and output streams.

Step 6: Use try and finally to reverse the contents of the file.

Step 7: Convert the contents using .ToUpper() method.

Step 8: Close the files when the operations are completed.

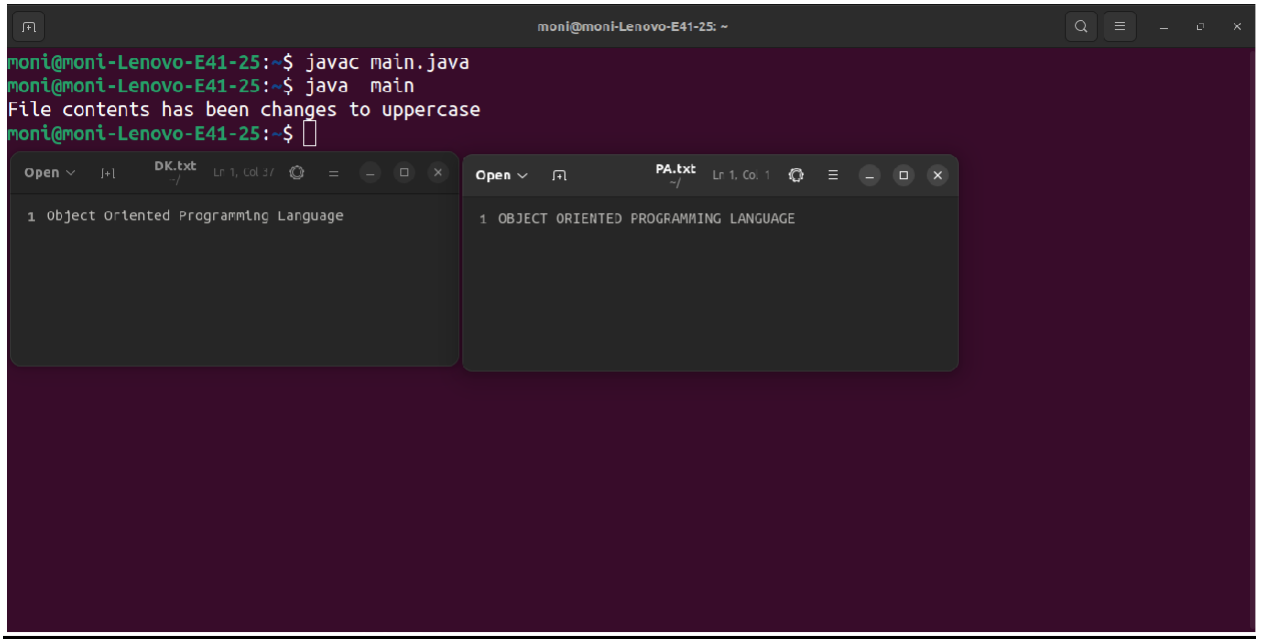
Step 9: Stop.

PROGRAM

```
import java.io.*;
import java.util.*;
class main
{
	public static void main(String args[]) throws IOException
	{
		FileInputStream fis= new FileInputStream("DK.txt");
		FileOutputStream fos = new FileOutputStream("PA.txt");
		try
		{
			int ch;
			String str="";
			while((ch=fis.read())!=-1)
			{
				str+=(char)ch;
			}
```

```
                    for(int i=0;i<str.length();i++)
                    {
                        fos.write(str.toUpperCase().charAt(i));
                    }
                    System.out.println("File contents has been changes to uppercase");
                }
                finally
                {
                    if(fis!=null)
                        fis.close();
                    if(fos!=null)
                        fos.close();
                }
        }
}
```

OUTPUT



RESULT

Thus, the java program for file content case conversion has been executed successfully

EX.NO.: 8	SWING APPLICATION
DATE:	

AIM:

To create a simple Java Swing application that displays a window containing a button and a label. When the button is clicked, the label text changes.

ALGORITHM:

Step1: Import the required Swing and event packages.

Step2: Create a class that extends JFrame.

Step3: Create a JLabel and a JButton.

Step4: Add the label and button to the frame.

Sep5: Register an action listener for the button.

Step6: When the button is clicked, change the label text.

Step7: Set the frame size, layout, and visibility.

Step8: Run the program.

PROGRAM:

```
import javax.swing.*;
import java.awt.event.*;

public class SwingApplication extends JFrame implements ActionListener {

    JLabel label;
    JButton button;

    SwingApplication() {
        // Create label
        label = new JLabel("Welcome to Java Swing");
        label.setBounds(50, 50, 200, 30);

        // Create button
        button = new JButton("Click Me");
        button.setBounds(50, 100, 100, 30);

        // Add action listener
        button.addActionListener(this);

        // Add components to frame
        add(label);
```

```
        add(button);

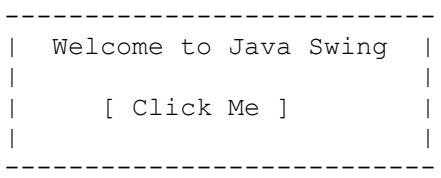
        // Frame settings
        setSize(300, 250);
        setLayout(null);
        setTitle("Swing Application");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        label.setText("Button Clicked!");
    }

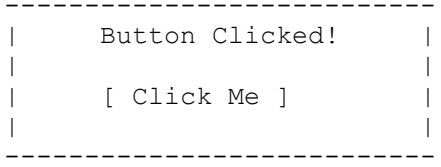
    public static void main(String[] args) {
        new SwingApplication();
    }
}
```

OUTPUT:

Initial Window



After Clicking the Button



RESULT:

The Java Swing application was successfully created and executed. A window containing a label and a button was displayed. When the button was clicked, the label text changed from "Welcome to Java Swing" to "Button Clicked!" demonstrating event handling in Swing.

EX.NO:9	DATABASE CONNECTION AND OPERATIONS (JDBC)
DATE:	

AIM:

To establish a database connection using JDBC and perform basic database operations such as inserting and retrieving records from a database table.

ALGORITHM:

Step1:Import the JDBC packages.

Step2: Load the JDBC driver.

Step3: Establish a connection to the database.

Step4: Create a statement object.

Step5: Create a table if it does not exist.

Step6: Insert records into the table.

Step7: Execute a SELECT query to retrieve records.

Step8: Display the retrieved records.

Step9: Close the database connection.

Step 10: End the program.

PROGRAM

```
import java.sql.*;

public class JDBCExample {
   public static void main(String[] args) {
      try {
         // Load JDBC Driver
         Class.forName("com.mysql.cj.jdbc.Driver");
```

```
        // Establish Connection
        Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/studentdb",
                "root",
                "password");

        Statement stmt = con.createStatement();

        // Create Table
        String createTable = "CREATE TABLE IF NOT EXISTS Student(" +
                             "id INT PRIMARY KEY, " +
                             "name VARCHAR(50))";
        stmt.executeUpdate(createTable);

        // Insert Record
        String insertQuery = "INSERT INTO Student VALUES(101,'John')";
        stmt.executeUpdate(insertQuery);

        System.out.println("Record inserted successfully.");

        // Retrieve Records
        ResultSet rs = stmt.executeQuery("SELECT * FROM Student");

        System.out.println("\nStudent Details:");
        while (rs.next()) {
            System.out.println("ID: " + rs.getInt("id") +
                               " Name: " + rs.getString("name"));
        }

        // Close Connection
        con.close();
      }
      catch (Exception e) {
        System.out.println(e);
      }
    }
}
```

OUTPUT:

Record inserted successfully.

Student Details:
ID: 101 Name: John

RESULT:

The database connection was established successfully using JDBC. The program created a table, inserted a record into the database, retrieved the stored data, and displayed it successfully.